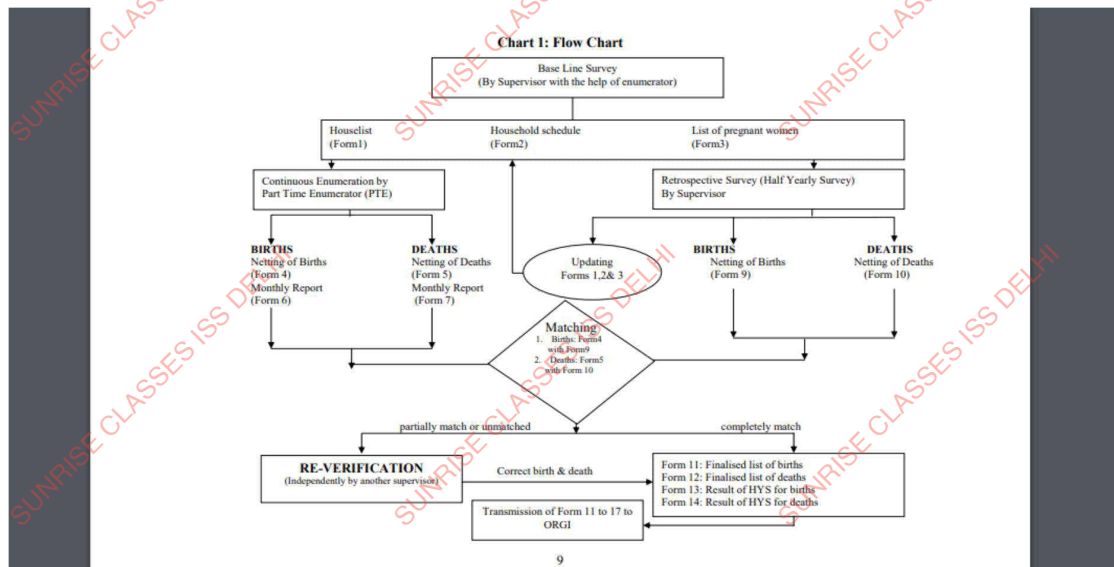


SRS (Sample Registration System)

- **objective-** to provide reliable estimates of birth rate, death rate and infant mortality rate at the natural division level for Major States and at the State level for Smaller States. It is a large-scale demographic survey conducted every year by Office of the Registrar General, India in all States/Union Territories.
- main components of SRS are:
 1. Base-line survey of the sample units to obtain usual resident population of the sample areas
 2. Continuous (longitudinal) enumeration of vital events pertaining to usual resident population by the enumerator
 3. Independent retrospective half-yearly surveys for recording births and deaths which occurred during the half-year under reference and up-dating the Houselist, Household schedule and the list of women in the reproductive age group along with their pregnancy status by the Supervisor;
 4. Matching of events recorded during continuous enumeration and those listed in course of half-yearly survey
 5. Field verification of unmatched and partially matched events.
 6. Filling of Verbal Autopsy Forms for finalized deaths.
- Process:

The field investigation consists of continuous enumeration of births and deaths in selected sample units by resident part-time enumerators, generally anganwadi workers and teachers; and an independent retrospective survey every six months by SRS supervisors. The data obtained by these two independent functionaries are matched. The unmatched and partially matched events are re-verified in the field and thereafter an unduplicated account of births and deaths is obtained.



- In SRS, 95% C.I. for Birth Rate, Death Rate and IMR are estimated using $BR \pm 2 SE(BR)$ and so on.
- Sampling Design:

Sample design adopted for SRS is a uni-stage stratified simple random sample without replacement except in stratum II (larger villages) of rural areas, where two stage stratification has been applied.

In urban areas, the categories of towns/cities have been divided into four strata based on the size classes in contrast to the six strata in the sampling frame prior to 2001.

Towns with population less than one lakh have been placed under stratum I, towns/cities with population one lakh or more but less than 5 lakhs under stratum II, towns/cities with population 5 lakh or more under stratum III and four metro cities of 5 Delhi, Mumbai, Chennai and Kolkata as separate strata viz. stratum IV. The sampling unit in urban area is a Census Enumeration Block. The Census Enumeration Blocks within each size stratum were ordered by the female literacy rate based on the Census 2011 data, and three equal size substrata were established. The sample Census Enumeration Block within each substratum was selected at random with equal probability. A simple random sample of these enumeration blocks have been selected within each sub-strata without replacement from each of the size classes of towns/cities in each State/Union Territory.

- SRS Bulletin (Oct 2021) Reference Year:2019

Parameters	Year 2019	Year 2018	Highest	Lowest
Birth Rate (Births per thousand pop)	19.7	20.0	Bihar(25.8)	A&N Islands(11.0)
Death Rate (Deaths per thousand pop)	6.0	6.2	Chattisgarh(7.3)	Delhi(3.2)
IMR (Infant deaths per thousand live births)	30	32	Madhya Pradesh(46)	Mizoram&Nagaland(3)

- Formulae Used

Crude birth rate	$\frac{\text{Number of live births during the year}}{\text{Mid-year population}} \times 1000$
Age-specific fertility	$\frac{\text{Number of live births in a particular age-group}}{\text{Mid-year female population of the same age-group}} \times 1000$
General fertility rate (GFR)	$\frac{\text{Number of live births in a year}}{\text{Mid-year female population in the age-group (15-49) years}} \times 1000$
Age-specific marital fertility rate (ASMFR)	$\frac{\text{Number of live births in a particular age-group}}{\text{Mid-year married female population of the same age-group}} \times 1000$

General marital fertility rate (GMFR)	$\frac{\text{Number of live births in a year}}{\text{Mid-year married female population in the age group (15-49) year}} \times 1000$
Crude death rate (CDR)	$\frac{\text{Number of deaths during the year}}{\text{Mid-year population}} \times 1000$
Age-specific mortality rate (ASMR)	$\frac{\text{Number of deaths in a particular age-group}}{\text{Mid-year population of the same age-group}} \times 1000$
Infant mortality rate (IMR)	$\frac{\text{Number of infant deaths during the year}}{\text{Number of live births during the year}} \times 1000$
Infant mortality rate comprises of two parts, viz. Neo-natal mortality rate and Post neo-natal mortality rate. The neo-natal mortality rate also comprises of two parts viz. Early neo-natal mortality rate and late neo-natal mortality rate. These are defined as:	
Neo-natal mortality rate (NMR)	$\frac{\text{Number of infant deaths of < than 29 days during the year}}{\text{Number of live births during the year}} \times 1000$
Early neo-natal mortality rate	$\frac{\text{Number of infant deaths of < than 7 days during the year}}{\text{Number of live births during the year}} \times 1000$

Late neo-natal mortality rate	$\frac{\text{Number of infant deaths of 7 days to < than 29 days during the year}}{\text{Number of live births during the year}} \times 1000$
Post neo-natal mortality rate (PNMR)	$\frac{\text{Number of infant deaths of 29 days to < than one year during the year}}{\text{Number of live births during the year}} \times 1000$
Peri-natal mortality Rate (PMR)	$\frac{\text{Number of still births and infant deaths of < than 7 days during the year}}{\text{Number of live births and still births during the year}} \times 1000$
Still birth rate	$\frac{\text{Number of still births during the year}}{\text{Number of live births and still births during the year}} \times 1000$

- Medically Certified Causes of Deaths (MCCD)

Following are eight leading cause-groups of deaths constituting around 87.2 per cent of total medically certified cause of deaths:

- ❖ Diseases of Circulatory System (32.7 per cent)
- ❖ Certain Infectious and Parasitic Diseases (9.8 per cent)
- ❖ Diseases of Respiratory System (9.7 per cent)
- ❖ Endocrine, Nutritional and Metabolic Diseases (5.7 per cent)
- ❖ Neoplasm (5.6 per cent)
- ❖ Certain Conditions Originating in the Perinatal Period (5.4 per cent)
- ❖ Injury, Poisoning and Certain other Consequences of External Causes (5.0 per cent).
- ❖ Symptoms Signs & Abnormal Clinical Findings "Not Elsewhere Classified" (13.3 per cent)